

Key

JAWAHARLAL NEHRU UNIVERSITY

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Mid-Term Exam

M.CA. (Winter Semester-2025)

Course Name: Computer Architecture

Course Code: CS 106

Maximum Marks: 40

Total Time: 2 hour

Q1. a) What is the difference between Computer Organization and Computer Architecture? Which parameters are directly related to processing power of the computer?

b) Amdahl's law is applicable to parallel processing or pipeline processing?

c) Write an assembly language program to show the difference between RISC and CISC Instruction set.

(4+2+4)

Q2. a) What are different types of addressing modes in 8086 microprocessor

b) A weather forecasting computation requires 40 Trillion floating point operations. The problem is processed in a Super computer that can perform 80 megaflops. How long will it take to perform these calculations?

10^6

c) A scientific simulation program has the following execution profile:

- 50% of the workload is fully parallelizable.
- 30% of the workload is partially parallelizable across at most 8 processors.
- The remaining 20% is completely serial.

You upgrade the system to a 32-processor machine. Compute the program's overall speedup on the 32-processor machine. Compute the utilization of the 32 processors (i.e., the fraction of total potential processing power used).

(3+3+4)

Q3. (a) Explain the important parameters that defines the performance of a Memory System? How can one increase the relative size in a Memory system? What is the 90/10 rule? What is the difference between Temporal and Spatial locality in memory system?

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b) Consider a CPU with an average CPI as 2.0. Assume the instruction mix is ALU-35%, Load-15%, Store-20% Branch-30%. Assume the cache miss rate of 5% and miss penalty of 60 cycles ($=t_{\text{miss}}$). Calculate the effective CPI for a unified L1 cache.

6/6

(5+5)

Q4 (a) What are the total external connections in 256×32 Memory system?

(b) Explain the difference between Direct, Full Associative and Set Associative cache mapping. Which one is preferred and why?

(c) Explain the Arithmetic pipeline for performing multiplication. Deduce the reservation Table and determine its MAL?

(2+3+5)

25×100